Stochastic

Probability, sampling & Monte Carlo. All in SQL.

Install & load

```
INSTALL stochastic FROM community;
LOAD stochastic;
```

01 One naming pattern

```
dist_<family>_<operation>(params..., x)
```

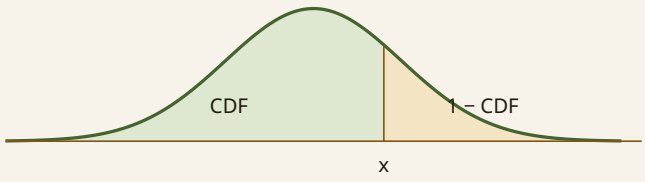
Distribution parameters come first. Add x for density or cumulative probability, p for a quantile, and neither for sampling or properties.

Operation suffix	Returns
`sample`	One random draw
`pdf`	Density / discrete mass[1]
`cdf`	Left tail: P(X ≤ x)
`cdf_complement`	Right tail: P(X > x)
`quantile`	Inverse CDF at p
`quantile_complement`	Inverse right tail at p
`log_pdf`	Natural log of PDF
`log_cdf`	Natural log of CDF
`log_cdf_complement`	Natural log of right tail
`hazard`	PDF / right tail
`chf`	−ln(right tail)

Properties: no trailing x or p

`mean`, `median`, `mode`, `variance`, `stddev`, `skewness`, `kurtosis`, `kurtosis_excess`, `range`, `support`. Availability depends on the family; `range` and `support` return two-element arrays.

02 Read the probability



The CDF is the shaded area left of x. Its complement is the area to the right. For continuous distributions, `pdf(x)` is a density, not P(X = x).

Tail arithmetic

For discrete counts: P(X ≥ k) = `cdf_complement(params..., k-1)`. Call the complement directly rather than subtracting a CDF from 1.

03 Normal: quantiles & CDFs

Parameters are mean and **standard deviation**.

```
SELECT dist_normal_quantile(0, 1, 0.95);
-- 1.6448536269514729
SELECT dist_normal_cdf(0, 1, 1.96);
-- approximately 0.975
```

04 Count probabilities

```
-- P(X ⩾ 5), Poisson(rate = 2.5)
SELECT dist_poisson_cdf_complement(2.5, 4);
-- 0.10882198108584878

-- P(X ⩽ 5), Binomial(n = 10, p = 0.3)
SELECT dist_binomial_cdf(10, 0.3, 5);
-- 0.9526510126000001
```

05 Generate → evaluate → average

Estimate π from uniform points in the square. Materialize draws so every expression reuses the same sampled coordinates.

```
WITH draws AS MATERIALIZED (
  SELECT
    dist_uniform_real_sample(-1, 1) AS x,
    dist_uniform_real_sample(-1, 1) AS y
  FROM range(100000)
)
SELECT 4.0 * AVG(
  (x*x + y*y ⩽ 1)::INTEGER
) AS pi_estimate
FROM draws;
```

Result varies by run and approaches π as the sample size grows.

Keep draws for reproducibility

`_sample` has no seed argument. Save draws with `CREATE TABLE ... AS SELECT ...` and reuse that table for repeatable analysis.

[1] `uniform_int` uses inclusive integer endpoints; see page 2. Discrete quantiles use family-specific rounding.

Full docs & examples ↗ Source ↗

Stochastic

The distribution directory

Use the family token in `dist_<family>_<operation>`. Parameters below are listed in call order; append `x` or `p` when the operation needs it.

06 Continuous families

Family token	Parameters / interpretation
`beta`	alpha, beta Positive shapes; proportions
`cauchy`	location, scale Heavy tails; no mean / variance
`chi_squared`	df Positive degrees of freedom
`exponential`	rate Mean waiting time = 1/rate
`extreme_value`	location, scale Gumbel; block maxima
`fisher_f`	df1, df2 Ratio of scaled variances
`gamma`	shape, scale Both positive; mean = shape × scale
`laplace`	location, scale Symmetric, sharper peak
`logistic`	location, scale Symmetric, heavier tails
`lognormal`	meanlog, sdlog Parameters of ln(X)
`normal`	mean, stddev Location and spread
`pareto`	scale, shape * Lower bound and tail shape
`rayleigh`	scale Nonnegative magnitudes
`students_t`	df Standard t; positive df
`uniform_real`	min, max min < max
`weibull`	shape, scale Both positive; failure times

* Pareto's runtime order is (scale, shape). Fixed v1.5 labels match this order. Older 234c881 labels incorrectly say (shape, minimum).

07 Discrete sampling families

Family token	Parameters / interpretation
`bernoulli`	prob A Boolean draw; 0 ≤ prob ≤ 1
`binomial`	trials, prob Successes in a fixed trial count
`geometric`	prob Failures before first success
`negative_binomial`	successes, prob Failures before target successes
`poisson`	rate Positive expected event count
`uniform_int`	min, max Integer draws; both ends included

Integer uniform: inclusive endpoints

Fixed v1.5 analytics match integer sampling. For bounds 1 to 6, PMF = 1/6, CDF at 3 = 0.5, mean = 3.5 and variance = 35/12. Older 234c881 uses incorrect continuous formulas and truncates moments.

08 Make a synthetic dataset

```
CREATE TABLE synthetic_users AS
SELECT
  i AS user_id,
  dist_normal_sample(170, 8)
    AS height_cm,
  dist_poisson_sample(3.5) AS sessions,
  dist_bernoulli_sample(0.12)
    AS converted
FROM range(10000) AS t(i);
```

Each row receives fresh draws. Bernoulli returns `BOOLEAN`; other discrete samplers return `BIGINT`; continuous samplers return `DOUBLE`.

09 Check the installed surface

```
SELECT function_name, parameters
FROM duckdb_functions()
WHERE function_name LIKE 'dist_normal_%'
ORDER BY function_name;
```

Use positive scales and standard deviations. Properties may be undefined at some parameter values; invalid inputs can raise SQL errors.

Full docs & examples ↗ Source ↗